

Carbon Black Pigment for Food Packaging Printing Inks

Technical Information TI 1453



Introduction

The safety of food is vital to all consumers and regulations governing the actual food products and packaging materials are well established and enforced. However, it was only in the last 20 years that any regulations relating to Carbon Black Pigments used for food packaging were issued – and initially only for plastics.

In the printing ink market, in the past, we have seen a very high growth rate for printed products for convenience foods and especially for the flexible packaging. We are now seeing this high growth rate also in the developing countries. Demand by consumers for more information on the packaging e. g. best before dates, nutrition information etc., all means higher ink consumption.

Although this is good news for the printing ink manufacturers, more and more are aware that some ink components are not suitable for food packaging. Due to this many ink manufacturers are working on special ink formulations suitable for food packaging.

The regulations in place for Europe are primarily concerned with the toluene extractables and the Benzo(a)pyrene content.

Orion Engineered Carbons can offer several products which are recommended for indirect food contact suitable for food packaging because they fulfill the requirements of the relevant regulations.

Relevant Regulations

The regulations mentioned in the introduction are listed below:

1) Commission Regulation (EU) No. 10/2011

“Carbon black”, FCM*-substance No 411, Ref. No. 42080 is listed as additive in Annex I “Union list” of Commission Regulation (EU) No. 10/2011 on plastic materials and articles intended to come into contact with food with the following specification:

- Primary particles of 10 – 300 nm which are aggregated to a size of 100 – 1200 nm which may form agglomerates within the size distribution of 300 nm – mm
- Toluene extractables: maximum 0.1 %, determined according to ISO method 6209
- UV absorption of cyclohexane extract at 386 nm: < 0.02 AU for a 1 cm cell or < 0.1 AU for a 5 cm cell, determined according to a generally recognised method of analysis
- Benzo(a)pyrene content: max 0.25 mg/kg Carbon Black

For FCM-substance No 411 the following restriction applies:

- Maximum use level of Carbon Black in the polymer: 2.5 % w/w



*) FCM = Food Contact Material

2) Swiss Ordinance 817.023.21

The Swiss Federal Department of Home Affairs (FDHA) recently adopted an amendment to the Ordinance of November 23rd, 2005 on materials and articles (SR 817.023.21) detailing certain provisions relating to packaging inks. Within this amendment, introduced on March 7th, 2008 and coming into force on April 1st, 2010, Article 26g details the requirement that only permitted substances should be used in the manufacture of packaging inks.

In amendment of SR 817.023.21 (dated February 1st, 2011), Annex 6, list II, part A, Carbon Black is listed to packaging inks. The purity specifications for pigments are defined in list III, item 9 (November 23rd, 2005).

3) French AVIS Séance du 07.11.1995

(encres et vernis pour l'impression des emballages)

Specification for "Carbon black"

Toluene extractables: max. 0.15 %

Benzo(a)pyrene content: $\leq 30 \mu\text{g/kg}$ Carbon Black

The French regulation AVIS séance du 07.11.1995 has the most stringent specification. However, OEC offers products which meet these requirements. Please refer to the table below:

Product Recommendations

Application	Recommendation*
Sheetfed	PRINTEX® 45 powder PRINTEX® 35 powder PRINTEX® Nature
Solvent based packaging	PRINTEX® 35 powder or beads SPECIAL BLACK 350 powder SPECIAL BLACK 535 beads PRINTEX® Nature
Water based packaging for cardboards	PRINTEX® 3 powder PRINTEX® 30 powder PRINTEX® Nature
Water based packaging for film	PRINTEX® 45 powder PRINTEX® 35 powder PRINTEX® Nature
UV curing systems	SPECIAL BLACK 350

* Valid only for grades manufactured in Germany

Recommendation is based on the physico chemical data and the resulting performance in the particular printing ink systems.

Product Safety Information

More information regarding the product safety status of our Carbon Black Pigments relating to the different requirements in the various countries can be obtained on request.



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